

PropX

AI/ML Engineer

Technical Assessment: Rooftop Detection & Attribute Extraction from Open Source Data

Second-round technical assessment

About this task

This is a practical take-home task that reflects the kind of work you would do with us. It is designed to take roughly one focused day of work, and you have 10 days from receiving it to submit.

We care more about the quality of your choices and your reasoning than about covering everything. Quality beats completeness: if you leave something out or make a simplifying assumption, just say so in your README.

What we evaluate

- (1) how you choose and justify your data source, (2) the quality of your roof detection and the attributes you extract, (3) your engineering and reproducibility, and (4) your reasoning, trade-offs and communication.

Some data sources you can explore. However, you can always look for more data sources for information fusion.

- i) data.gv.at OGD / WMS [Stadtplan Wien](#)
- ii) Sentinel-2 (<https://dataspace.copernicus.eu>)
- iii) surface-temperature / thermal layers for insulation
- iv) street view images
- v) Mapillary data
- vi) Cesium ion

Background

A recurring problem when working with 3D building data is that most buildings don't have detailed digital model. A lot of useful information can instead be derived automatically from openly available imagery and geodata. This task focuses on one such problem: understanding a building's roof from open sources.

The task

For a small set of real buildings in a city of Vienna, build a component that, from open imagery/geodata, detects each building's roof and extracts as many useful roof attributes as you can reliably justify, each with a confidence score.

Pick 8–10 buildings in a small area and list them, so your results are self-contained and comparable.

1) Choose your data source(s) and justify it

Part of the task is choosing wisely. You may use any open sources. Common options include:

- Aerial orthophotos (high-resolution, top-down).
- Oblique / “bird’s-eye” aerial imagery (angled views).

- Optical satellite imagery (e.g. Copernicus Sentinel-2).
- 3D city meshes / photorealistic 3D tiles.
- Street-level or 360° imagery.
- Open building footprints (e.g. OpenStreetMap or open government data portals such as data.gv.at).

Required: clearly explain which imagery/source(s) you selected for roof detection and why — resolution, coverage, obstruction, licence/cost, and the trade-offs you weighed. If you tried a source and rejected it, tell us why.

2) Extract roof attributes

From your chosen source, detect the roof and extract as many of the following as you can reliably support, each with a per-attribute confidence score:

- Roof outline / polygon
- Roof area.
- Roof type (e.g. flat, pitched, hipped, complex).
- Roof material or surface (e.g. tiled, metal, flat/gravel, green roof, glass).
- Anything else you can justify (e.g. solar panel, superstructures, visible condition).

Required: state which attributes your chosen source allowed you to extract and which it did not, and why. We are specifically interested in your judgement about what is and isn't recoverable from a given source and how you merge data from different data sources to complete the building information.

Provide your results as a structured file (e.g. JSON), one record per building. A suggested shape:

```
{
  "building_id": "...",
  "source_used": "...",
  "roof": {
    "polygon": [[lon, lat], ...], "area_m2": 0.0,
    "type": "...", "material": "...", "orientation": "...", "slope_deg": 0.0
  },
  "confidence": { "area": 0.0, "type": 0.0, "material": 0.0, "...": 0.0 },
  "notes": "..."
}
```

3) Design & reasoning write-up (about one page)

In your own words, cover:

- Why you chose your source(s) and the main trade-offs.
- Which roof attributes that source made possible, and which it didn't and why.
- How you would align/geolocate the imagery to the correct building, and how you would scale this from 10 buildings to many thousands.
- How you represent and would use the confidence scores.

Deliverables

- A Git repository (GitHub or GitLab), public or shared with us (shaily.gandhi@gmail.com).

- A README written as if for a production service: what the problem/solution is, how to install and run it (ideally one command), the tools/models/APIs you used, your key design choices, trade-offs, and what you would do with more time.
- Your structured inputs and results file formats and data cleaning (e.g. outputs/roof_attributes.json).
- 3–5 images showing detected roofs overlaid on the imagery.
- The one-page design & reasoning write-up .

How we evaluate

Your submission is scored on four areas:

Area	What we look for
Source selection & justification	A sensible choice of source(s) for the goal, with clear reasoning about resolution, coverage, obstruction, licence and trade-offs.
Roof detection & attribute extraction	Correct, plausible roof detections; a useful set of attributes; honest judgement about what each source supports; sensible confidence scores.
Engineering & reproducibility	Clean, readable code that runs from your README; good repo hygiene; you used your time well.
Reasoning, trade-offs & communication	A clear write-up and README; honest limitations; depth appropriate to your level (architecture/scaling for senior roles).

How to submit

- Reply to this email with your repository link (and access if it is private) within 10 days.
- Use any language, tools, models or AI assistants you like just tell us what you used and note any assumptions.
- Questions are welcome at any point just reply to this email.